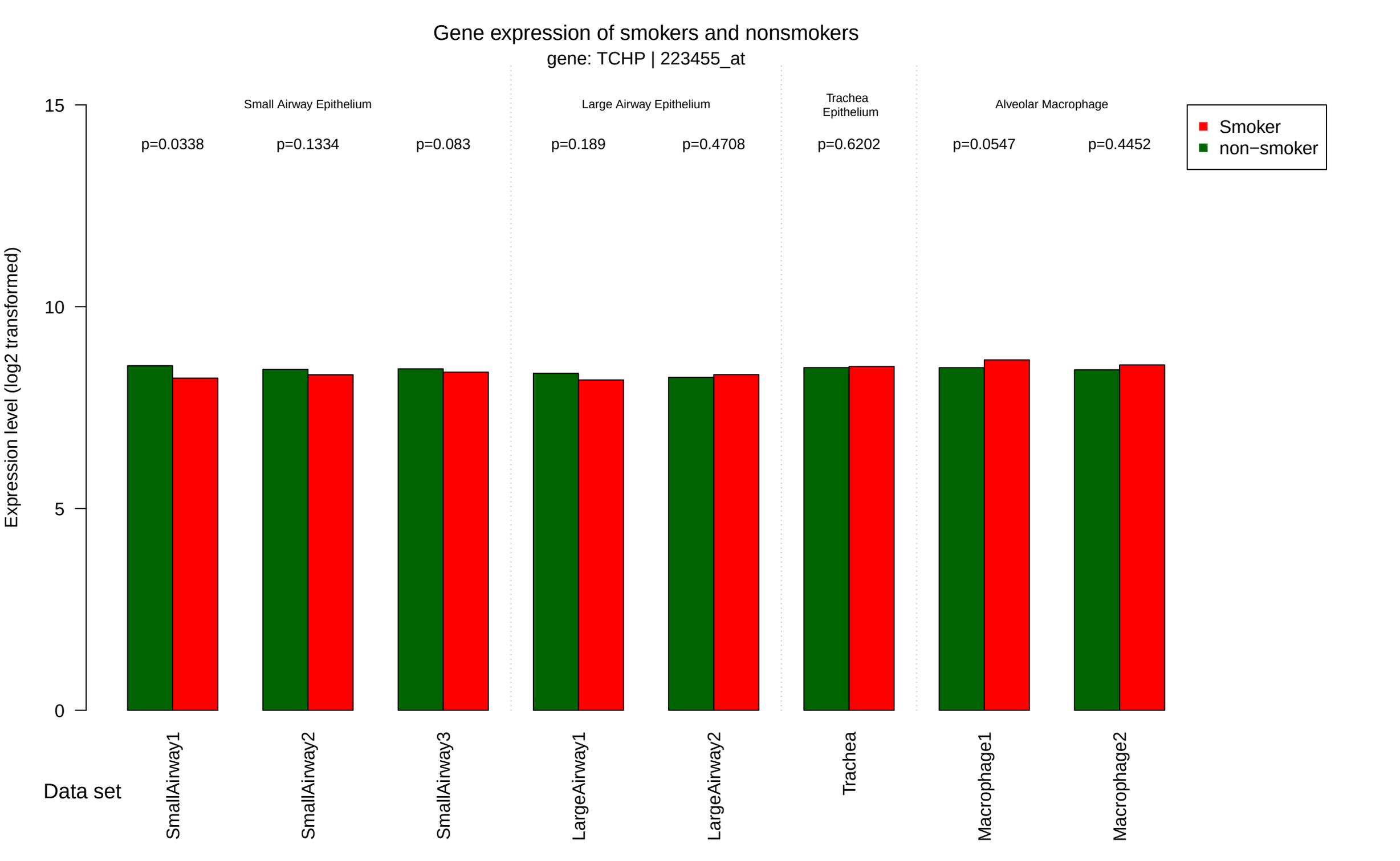
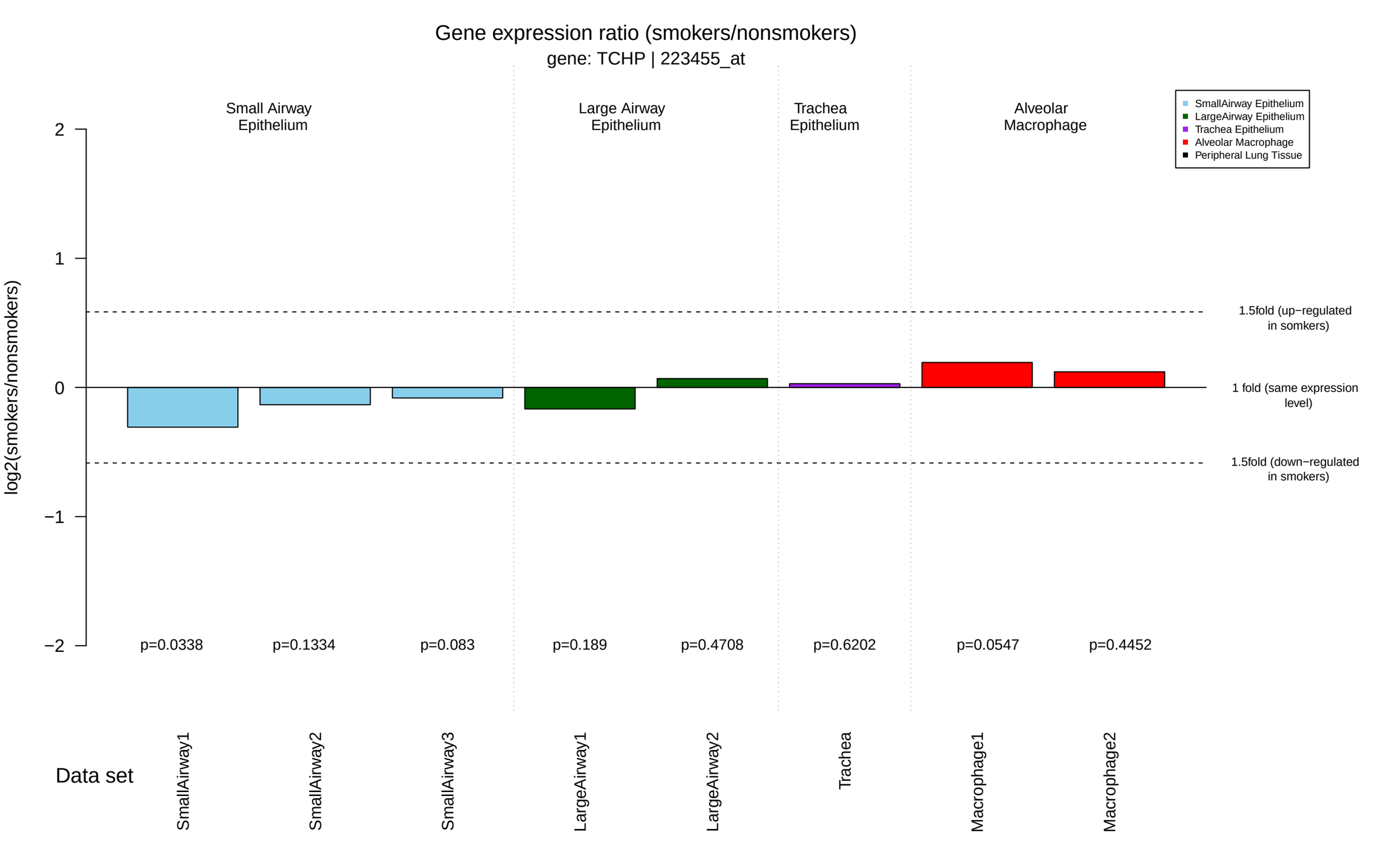
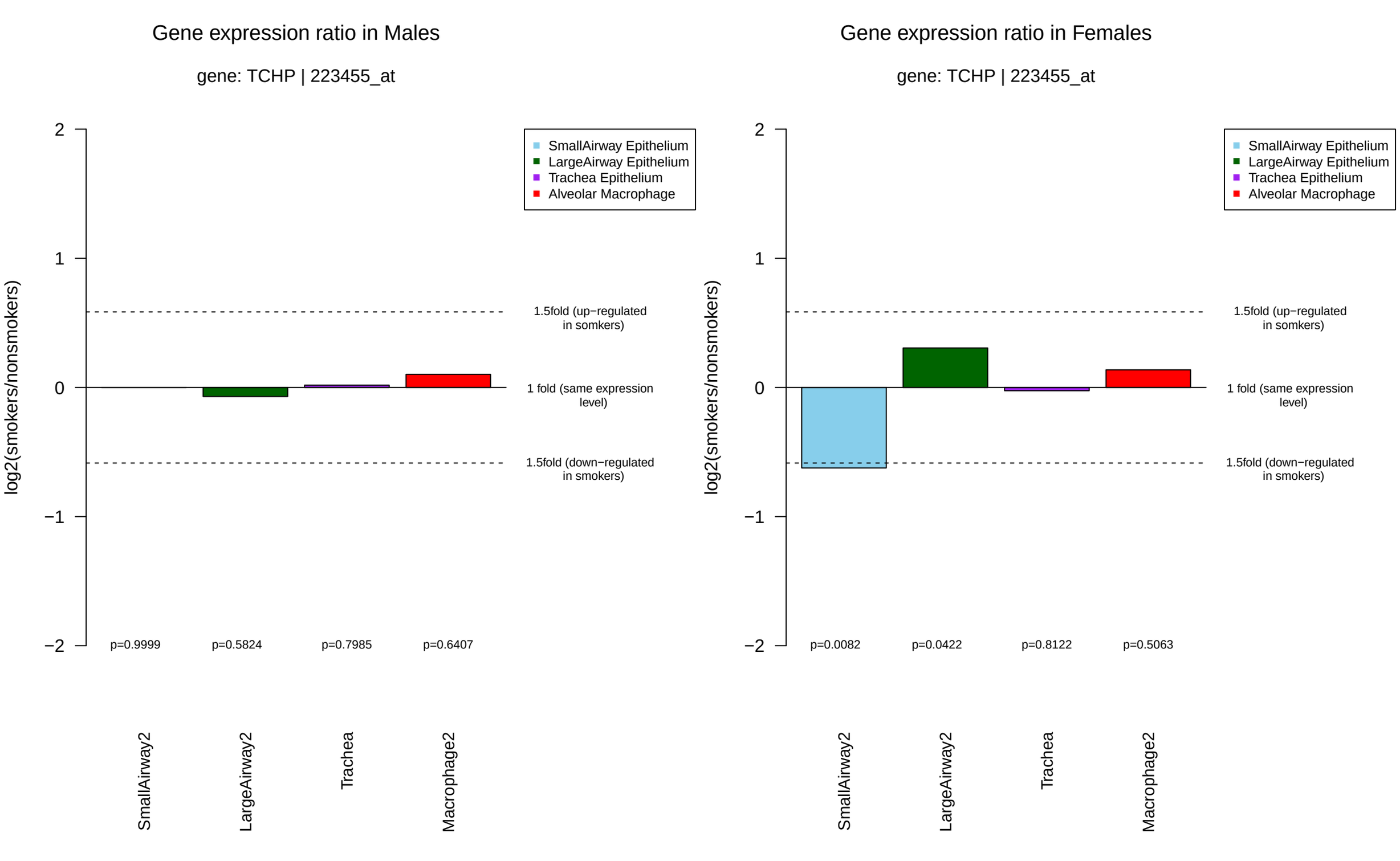
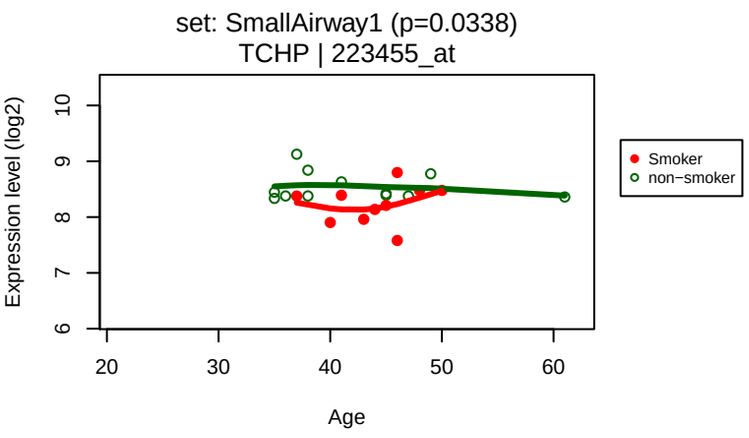
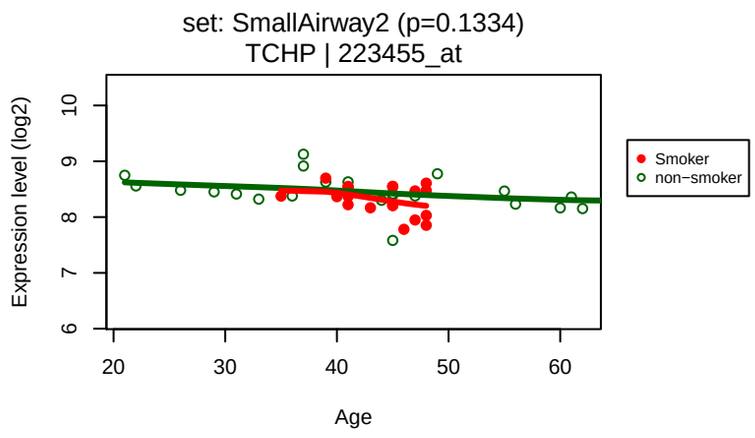
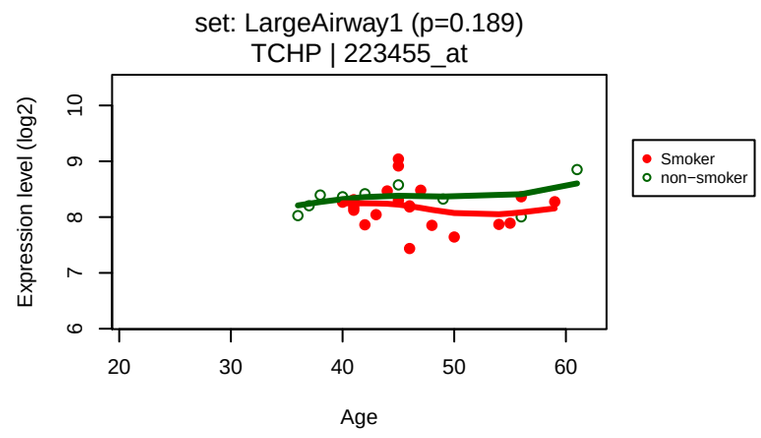

Gene expression of smokers and nonsmokers
gene: TCHP | 223455_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0338
p=0.1334
p=0.083
p=0.189
p=0.4708
p=0.6202
p=0.0547
p=0.4452
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

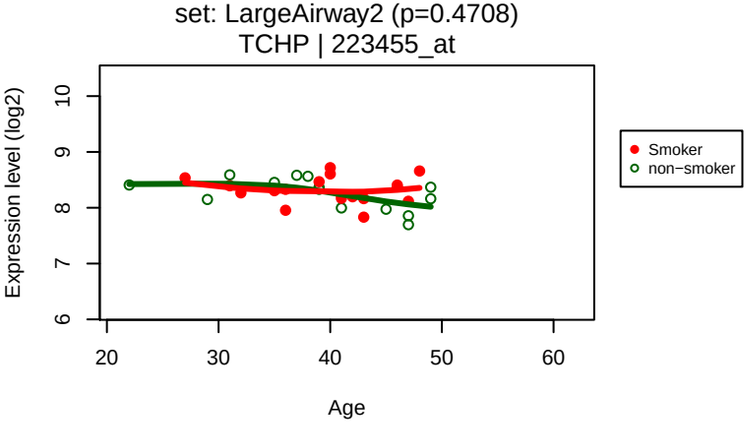

Gene expression ratio (smokers/nonsmokers)
gene: TCHP | 223455_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0338
p=0.1334
p=0.083
p=0.189
p=0.4708
p=0.6202
p=0.0547
p=0.4452
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

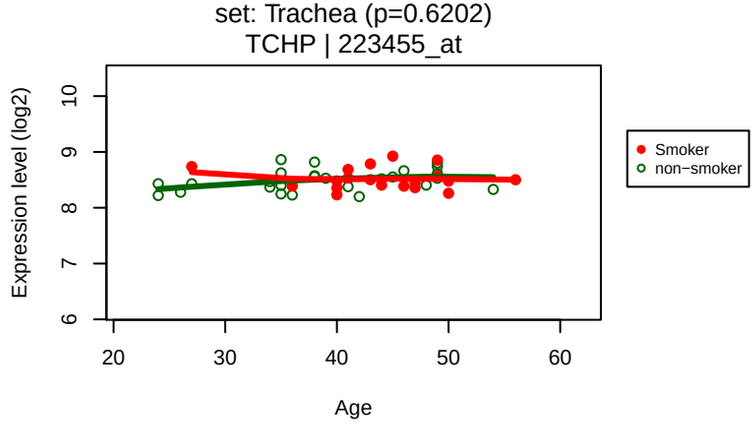

Gene expression ratio in Males
gene: TCHP | 223455_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9999
p=0.5824
p=0.7985
p=0.6407
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TCHP | 223455_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0082
p=0.0422
p=0.8122
p=0.5063
SmallAirway2
LargeAirway2
Trachea
Macrophage2

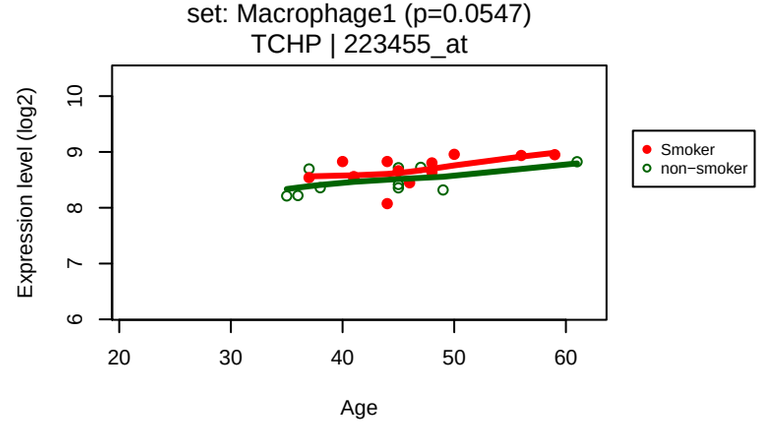

set: SmallAirway1 (p=0.0338)
TCHP | 223455_at
Expression level (log2)
Age
Smoker
non−smoker

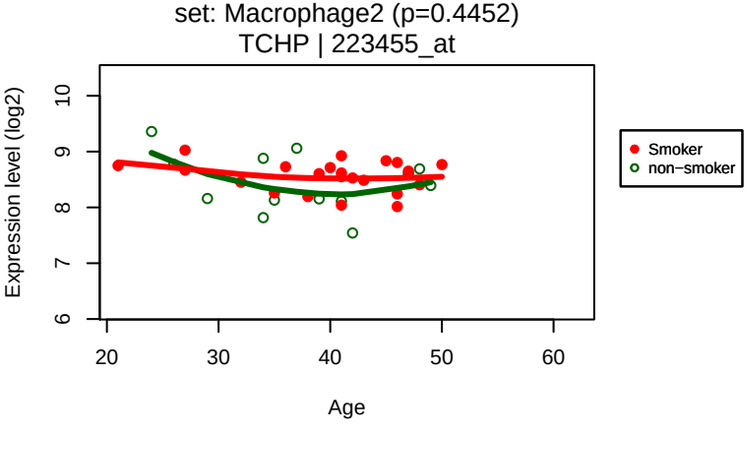

set: SmallAirway2 (p=0.1334)
TCHP | 223455_at
Expression level (log2)
Age
Smoker
non−smoker

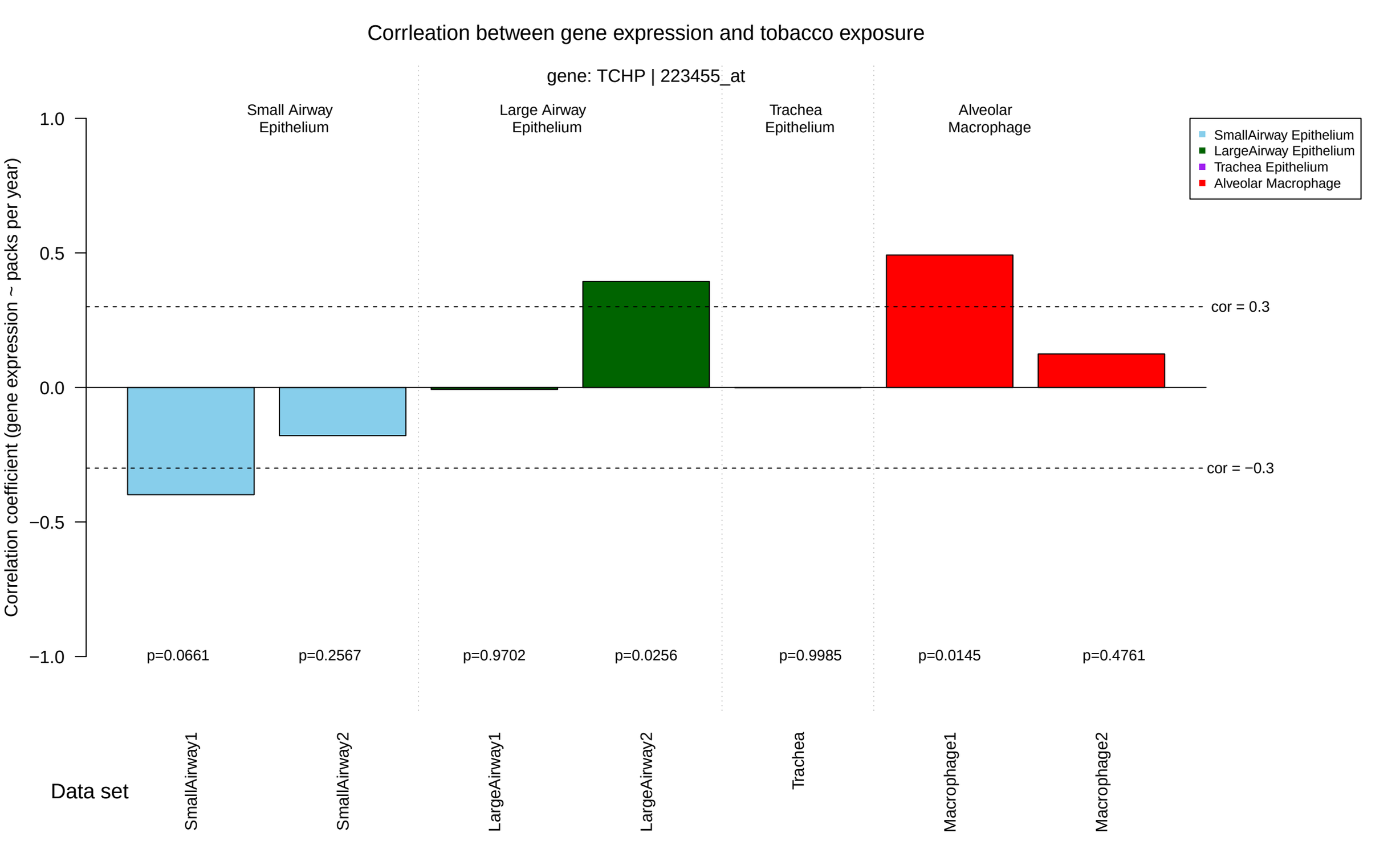

set: LargeAirway1 (p=0.189)
TCHP | 223455_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4708)
TCHP | 223455_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6202)
TCHP | 223455_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0547)
TCHP | 223455_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4452)
TCHP | 223455_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TCHP | 223455_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.0661
p=0.2567
p=0.9702
p=0.0256
p=0.9985
p=0.0145
p=0.4761
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2